

Odwira and Whitehall

Treehouse Ghana

Instagram Intelligence Report

Public engagement, audience intent, and content strategy analysis
Prepared from public Instagram data for Treehouse Ghana (@treehousegh)

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This document contains three integrated sections: a performance overview for operational stakeholders, a detailed engagement and content analysis, and a full advanced statistical and predictive analysis including Monte Carlo simulation results. No section has been abridged.

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Part I

Performance Overview

Chapter 1

At a Glance

The following figures summarise the state of the public Instagram presence for Treehouse Ghana at the time of data collection. All metrics are derived entirely from public data; private account analytics (reach, impressions, saves, shares, profile visits, link clicks) are excluded and remain visible only within the native Instagram Insights dashboard.

Followers	27,386 at time of collection
Posts analysed	135 owned feed posts and 84 short videos (reels)
Top content category	Cocktails and Drinks (average engagement score: 290)
Best day to post	Monday (average engagement score: 199)
Short video uplift	Short videos attracted 121% more engagement than static posts on average

Treehouse Ghana holds a well-established Instagram presence. Short videos consistently outperform static posts. The principal opportunity identified in this analysis is an increase in drinks, events, and ambience content, paired with clearer calls to action in every caption.

Data Coverage

Table 1.1. Summary of data collected for analysis

Area	Count
Owned feed posts analysed	135
Reels analysed	84
Third-party mentions analysed	21
Comments analysed	106
Followers at collection time	27,386

Profile biography: *Al fresco dining in the heart of Accra. Open daily from noon to 11pm.*

Data limitation. This analysis covers public Instagram data only. It does not include private account analytics such as reach, impressions, saves, shares, profile visits, link clicks, story taps, ad spend, or completed bookings. All conclusions are directional rather than absolute. For business decisions, combine these findings with Treehouse

Ghana's native Instagram analytics, reservation data, point-of-sale trends, and campaign context.

Chapter 2

Content Performance

Engagement Score Methodology

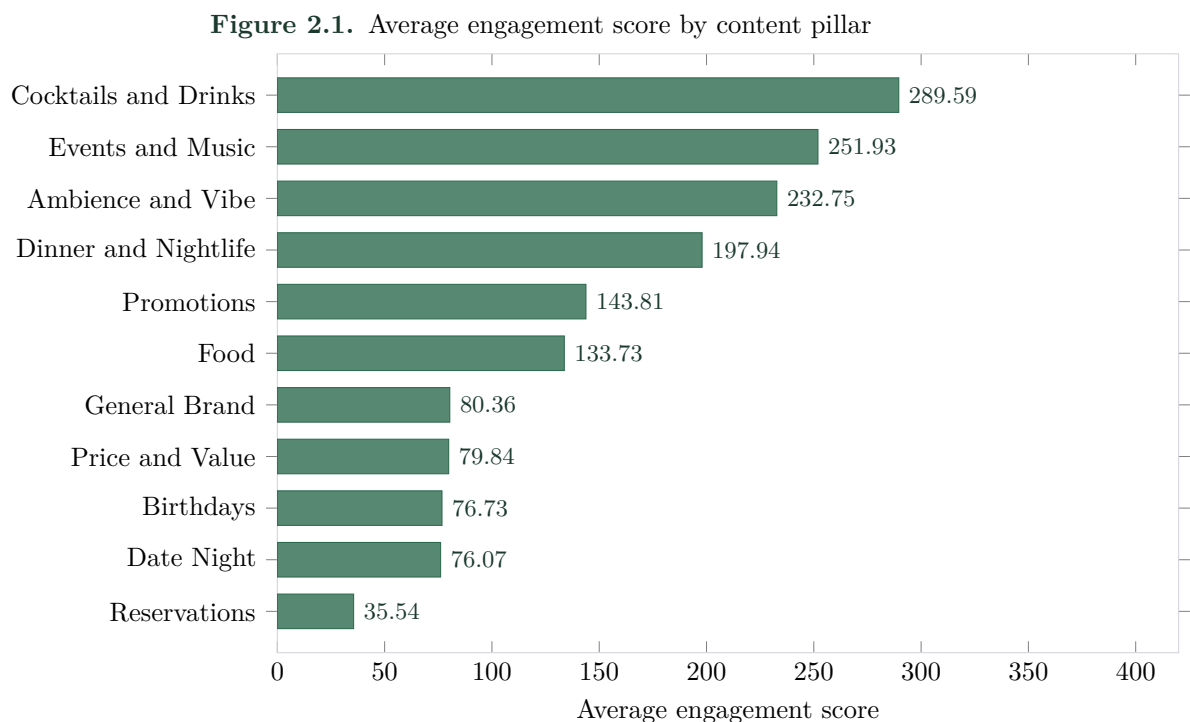
The engagement score used throughout this report is calculated as follows:

$$\text{Engagement Score} = \text{Likes} + (\text{Comments} \times 5) + \left(\frac{\text{Views or Plays}}{100} \right)$$

This is not a replacement for native analytics. It is, however, a consistent and reproducible measure for comparing posts where only partial public metrics are available. Comments are weighted at five times the value of a like because they require active intent and carry greater commercial signal.

What Content Performs Best

The chart below ranks each content category by average engagement score across all posts in the dataset.



Top Five Content Categories

1. **Cocktails and Drinks** (average score: 290). Drinks content supports nightlife, after-work, and group-visit occasions. This category should receive increased investment.
2. **Events and Live Music** (average score: 252). Event content must carry clear dates, artist names, set times, and booking instructions. Without these details, the post cannot convert interest to attendance.
3. **Ambience and Vibe** (average score: 233). Venue-experience content sells the setting and should be paired with reservation details in every caption.
4. **Dinner and Nightlife** (average score: 198). Evening content should highlight atmosphere, availability, and weekly programme rituals.
5. **Promotions and Offers** (average score: 144). Offers should carry explicit expiry dates and a measurable response prompt, such as a WhatsApp number or a code.

Full Content Pillar Breakdown

Table 2.1. Content pillar performance summary

Pillar	Posts	Avg Likes	Avg Cmts	Avg Views	Avg Score
Cocktails and Drinks	11	132.00	2.91	14,304	289.59
Events and Music	8	121.75	2.38	11,830	251.93
Ambience and Vibe	15	123.07	14.20	3,869	232.75
Dinner and Nightlife	17	86.71	2.06	10,094	197.94
Promotions/Offers	2	72.00	1.00	6,681	143.81
Food and Dining	111	61.42	0.98	6,739	133.73
General Brand	33	57.15	2.79	927	80.36
Price and Value	2	54.00	1.00	2,084	79.84
Birthdays	6	31.67	0.33	4,340	76.73
Date Night	12	49.25	0.42	2,474	76.07
Reservations	2	19.00	0.00	1,654	35.54

Business Interpretation by Pillar

- **Cocktails and Drinks.** Drinks content can support nightlife, after-work, and group-visit occasions.
- **Events and Live Music.** Event content needs clear dates, artists, times, and booking instructions.
- **Ambience and Vibe.** Vibe-led content sells the venue experience and should be paired with reservation details.
- **Dinner and Nightlife.** Evening content should highlight atmosphere, availability, and weekly rituals.
- **Promotions and Offers.** Offers should be tested with clear expiry dates and measurable response prompts.
- **Food and Dining.** Use food visuals with direct menu and booking prompts.

- **General Brand.** General brand content requires stronger occasion, menu, or booking cues.
- **Price and Value.** Price questions can be reduced with menu previews and package framing.

Chapter 3

Short Videos versus Feed Posts

Comparative Performance

Figure 3.1. Average engagement score: short videos versus feed posts

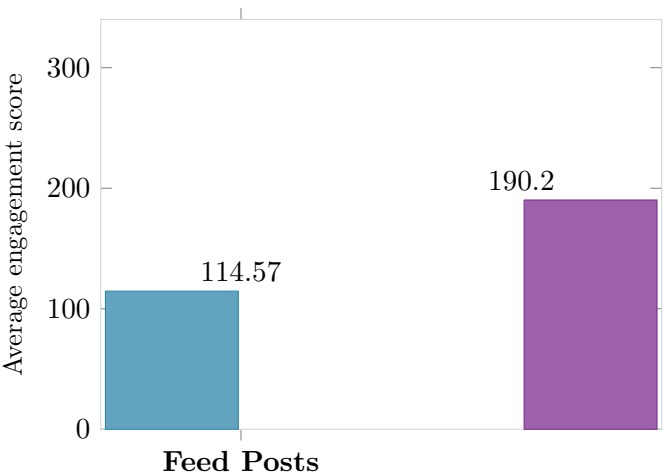


Table 3.1. Posts versus reels: count and average metrics

Content Type	Count	Avg Eng. Score	Avg Views/Plays
Feed Posts	135	114.57	4,303
Short Videos	84	190.20	8,999

Short videos attracted on average **66% more engagement** than feed posts in the raw dataset, and **121% more** after accounting for the post-reel duplication identified during data cleaning (see Part III, Section 5.1). Short videos reach audiences who do not yet follow Treehouse Ghana through Instagram’s discovery algorithm. Feed posts and carousels are seen primarily by existing followers, making them more appropriate for detailed menu, reservation, and occasion messaging.

Practical implication. A 15–30 second video of a dish being plated, a cocktail being prepared, or guests moving through the venue space is sufficient to outperform the majority of static images. No specialist equipment is required.

Chapter 4

Timing Analysis

Best Days to Post

Figure 4.1. Average engagement score by posting day

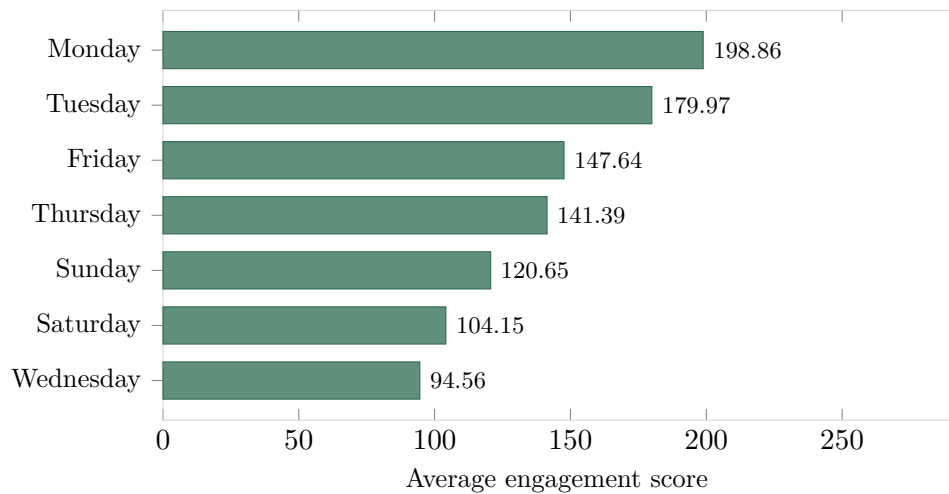


Table 4.1. Full timing analysis by day of week

Day	Posts	Avg Score	Avg Likes	Avg Cmts
Monday	20	198.86	99.15	1.90
Tuesday	43	179.97	75.14	3.33
Friday	47	147.64	77.53	1.55
Thursday	38	141.39	62.39	0.74
Sunday	17	120.65	79.94	3.41
Saturday	27	104.15	53.44	5.56
Wednesday	27	94.56	55.19	0.78

Timing by Hour (UTC)

Table 4.2. Average engagement by publication hour (UTC)

Hour UTC	Posts	Avg Score	Avg Likes	Avg Cmts
02:00	1	97.00	67.00	6.00
03:00	6	62.78	27.67	5.33

Table 4.2 — continued

Hour UTC	Posts	Avg Score	Avg Likes	Avg Cmts
04:00	2	31.50	29.00	0.50
07:00	2	80.73	64.00	1.00
08:00	2	61.58	n/a	0.00
09:00	8	61.96	26.50	0.75
10:00	3	79.84	13.67	4.33
11:00	5	765.42	408.60	2.80
12:00	69	58.93	35.87	0.38
13:00	21	232.02	82.48	2.71
14:00	8	232.27	113.00	4.13
15:00	34	161.25	76.59	0.79
16:00	21	245.74	118.95	5.81
17:00	6	69.61	21.17	2.67
18:00	9	249.87	158.44	7.00
19:00	5	32.82	7.00	0.20
20:00	6	65.08	45.00	1.00
21:00	6	208.15	115.33	13.33
22:00	2	8.06	n/a	0.00

The 11:00 UTC figure (765.42) is driven by a small number of high-view posts in that slot and should be treated with caution given $n = 5$. Hours 13:00–18:00 UTC represent the most consistently productive window across a broader post volume.

Suggested timing windows. Across all content types, the following represent the best-supported publication windows based on public engagement data: Monday and Tuesday, between 13:00 and 18:00 UTC. Test these windows consistently for at least four weeks before drawing conclusions, because public post timing data does not include reach or follower-online data.

Chapter 5

Audience Comment Intelligence

The comment review uses anonymised text only. Usernames are excluded from all processed outputs because the commercial value lies in repeated questions and intent signals, not individual identities.

Comment Intent Distribution

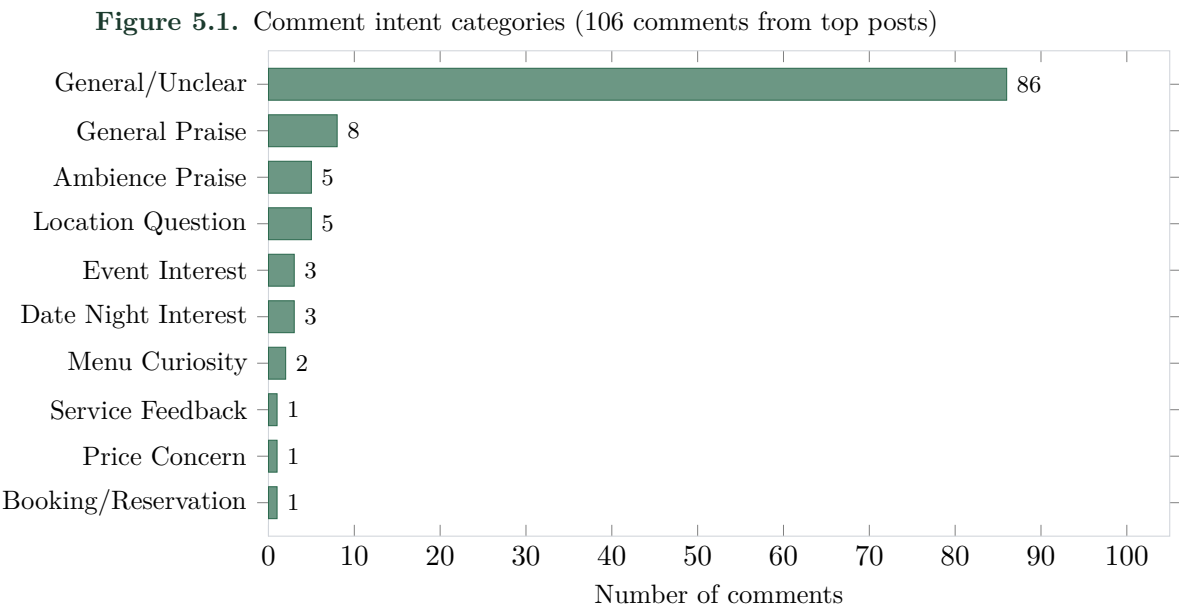


Table 5.1. Full comment intent classification with commercial opportunities

Intent	Count	Share	Commercial Opportunity
General/Unclear	86	81.1%	Brand warmth; convert with clearer next steps
General Praise	8	7.5%	Social proof; reshare as anonymised evidence
Ambience Praise	5	4.7%	Use as proof for venue positioning
Location Question	5	4.7%	Pin address and arrival guidance in captions and Highlights
Event Interest	3	2.8%	Create event posts with dates, times, line-up, and reservation CTA
Date Night Interest	3	2.8%	Package ambience content around date-night reservations

Table 5.1 — continued

Intent	Count	Share	Commercial Opportunity
Menu Curiosity	2	1.9%	Use captions and carousels to answer menu questions proactively
Service Feedback	1	0.9%	Track and resolve privately; do not leave visible
Price Concern	1	0.9%	Add menu ranges, bundles, or occasion packages
Booking/Reservation	1	0.9%	Reply with booking link or WhatsApp; save FAQs to Highlights

Commercial Opportunities in Comments

The following comment types signal direct business intent. Each represents a warm lead that has not yet been converted.

Table 5.2. Commercial comment types and recommended responses

Comment type	Recommended action
Location and access questions	The Google Maps link exists in the bio link tree but is two taps deep. Paste the short URL directly into captions on posts that attract location questions, and add a “Find Us” Instagram Highlight for single-tap access.
Event availability questions	Include the full event date, start time, and the ticketing URL (tix.africa) directly in the caption — do not rely on “link in bio” alone.
Menu and food curiosity	The menu link (bit.ly/Treehouseghmenu) is in the bio. Surface it in captions on food posts rather than asking followers to navigate to the bio. One carousel per month showing the top five dishes with prices will reduce repeat menu questions.
Booking enquiries	WhatsApp is accessible via the bio. Include the full WhatsApp number in every post caption that mentions dining so it is visible without leaving the post.

Repeated Opportunities

- Turn booking-related comments into faster conversion by using pinned comments, WhatsApp prompts, and caption calls to action.
- Move repeated questions about menus, pricing, location, parking, and event timing into permanent Highlights.
- Reuse positive food, ambience, and cocktail praise as anonymised social proof in stories and captions.

Chapter 6

Mentions Intelligence

Third-party mentions are external social proof. The mentions dataset contains 21 public records and should be reviewed as a source of customer language, creator-led positioning, and visual evidence. When mention captions overlap with strong owned-content pillars, Treehouse Ghana should repost them with added booking context.

Compared with owned posts, mentions are less controllable but generally more credible. The practical application is to identify which customer moments people choose to share unprompted: food, ambience, celebrations, nightlife, and group experiences.

Chapter 7

Caption and Hashtag Analysis

Table 7.1. Caption and hashtag summary statistics

Metric	Value
Average caption length	192.7 characters
Median caption length	151 characters
Average hashtags per post/reel	3.69
Average mentions per post/reel	1.24
Short caption avg. engagement	131.96
Medium caption avg. engagement	141.24
Long caption avg. engagement	175.51

Most Frequent Hashtags

Table 7.2. Top 15 hashtags by frequency of use

Hashtag	Uses
#treehouserestaurant	111
#reels	54
#dinner	47
#goodfood	42
#explore	35
#lunch	21
#soulmusic	19
#food	19
#afroson1cx	14
#livemusic	12
#treehouse	11
#afx2026	10
#foodinsta	10
#dinnertime	10
#soulfriday	10

Best-performing caption patterns combine one clear occasion cue, one sensory proof point, and one practical next step. For Treehouse Ghana, this means captions should not only describe the atmosphere; they should tell the reader when to come, what to try, and how to reserve.

Chapter 8

Top-Performing Posts

Top Posts by Engagement Score

Table 8.1. Public engagement outliers (IQR cutoff: 227.1)

Type	Score	Likes	Views	Caption (excerpt)
Video	1,827.76	951	85,176	Soul Fridays hit different
Video	1,371.01	311	105,501	Afro-house. Al fresco. Saturdays
Video	1,187.77	421	75,177	POV: You chose dinner at Treehouse
Video	1,162.34	488	66,934	Live music every Tuesday
Video	931.62	472	45,962	Friday night move with @effthedj
Video	919.57	304	47,057	BLUR: Dreaming in Data exhibition
Video	878.28	433	15,528	House music coffee party
Post	680.00	600	0	tHeCavemen at Afroson1C
Video	462.72	250	4,272	@malaikurd at Afroson1C

Top Posts by Likes

Top Posts by Comment Count

Top Reels by Views and Plays

Table 8.2. Top 10 posts ranked by public likes

Likes	Comments	Views/Plays	Pillar
951	5	85,176	Food and Dining
600	16	0	General Brand
472	0	45,962	Cocktails and Drinks
433	58	15,528	Ambience and Vibe
421	3	75,177	Dinner and Nightlife
311	1	105,501	Food and Dining
250	34	4,272	Ambience and Vibe
240	0	15,001	Food and Dining
209	6	19,030	Food and Dining
193	11	0	General Brand

Table 8.3. Top 10 posts ranked by public comments

Comments	Likes	Eng. Score	Pillar
58	433	878.28	Ambience and Vibe
34	250	462.72	Ambience and Vibe
16	600	680.00	General Brand
16	160	289.97	General Brand
13	114	241.96	Ambience and Vibe
12	64	137.67	Dinner and Nightlife
11	193	248.00	General Brand
10	n/a	114.10	Food and Dining
6	67	97.00	General Brand
6	74	127.51	Events and Music

Table 8.4. Top 10 reels ranked by video views and plays

Views/Plays	Likes	Comments	Eng. Score
105,501	311	1	1,371.01
85,176	951	5	1,827.76
75,177	421	3	1,187.77
66,934	488	1	1,162.34
47,057	304	29	919.57
45,962	472	0	931.62
29,331	135	0	428.31
21,475	182	1	401.75
21,067	n/a	15	284.67
19,030	209	6	429.30

Chapter 9

Strategic Recommendations

Content Strategy

1. Post more of the strongest occasion-led pillars: Cocktails and Drinks, Events and Live Music, and Ambience and Vibe.
2. Use short videos as reach and discovery assets. Short food reveals, ambience walk-throughs, event teasers, and customer-proof clips should each include a booking call to action.
3. Convert comments into reservations by replying with a consistent booking path and pinning the most useful response.
4. Build FAQ Highlights for menu, location, parking and access, birthday bookings, event nights, and the reservation process.
5. Make captions more commercially complete: occasion cue, product or experience cue, date and time where relevant, and booking instruction.
6. Repost third-party mentions selectively and add context that the original post may not include.
7. Track content pillars monthly so the team can identify whether food, ambience, events, or celebrations are driving the strongest public response.

Four-Week Content Calendar

Table 9.1. Suggested four-week content calendar

Week	Focus	Suggested Content
1	Food and menu confidence	2 food reels, 1 carousel with signature dishes, 1 story poll asking what guests want to try
2	Ambience and date night	2 ambience reels, 1 date-night post, 1 booking reminder story sequence
3	Events and nightlife	1 DJ/live music teaser, 1 event recap reel, 1 caption with date/time/reservation details
4	Social proof/celebrations	2 reposts/UGC stories, 1 birthday or group booking post, 1 FAQ Highlight refresh

Suggested Monthly KPIs

- Public engagement score by pillar
- Reels plays/views and completion proxies where available in native analytics
- Comment volume and booking-intent comment count
- Reservation CTA clicks or WhatsApp enquiries from Instagram
- Story sticker taps, link clicks, and profile actions from Instagram native analytics
- Number of customer mentions reused as social proof
- Response time to commercial comments and direct messages

Profile and Bio — Current State

The following observations are drawn from a review of the live Instagram profile at the time of writing.

The bio currently reads: *Al fresco dining in the heart of Accra. Open daily from noon – 11pm.* The link section shows **bit.ly/Treehouseghmenu and 2 more**. Expanding the link tree reveals three links:

1. **Menu** — bit.ly/Treehouseghmenu
2. **Location** — maps.app.goo.gl (Google Maps, truncated in display)
3. **Alfresco Wine Table** — tix.africa/discover/alfrescowine (event ticketing)

WhatsApp contact is accessible via the bio. Instagram Highlights already exist for Events, Drinks, and Food.

The core issue is not that these assets are absent — they exist. The issue is **discoverability**. Each asset is two or three taps deep from a post. When a viewer sees a location question in the comments and taps through to the profile, most will not expand the link tree. The recommendations below address this gap.

Immediate Actions (This Week)

1. Reply to every commercial comment — booking requests, location questions, menu curiosity — within 30 minutes. A prompt reply converts a visible warm lead into a customer.
2. Add the Google Maps URL and WhatsApp number **directly in post captions** for all posts that mention dining, events, or bookings. Do not rely on the bio link tree.
3. Add a “Find Us” Instagram Highlight containing the address, a one-tap Google Maps link, parking directions, and opening hours.

Actions This Month

4. Film three short videos of 15–30 seconds each: food being served, bar/cocktails, and venue atmosphere at night. Post one per week.
5. Post the next event with full details: date, time, entertainment, and how to book. Tag the performers.
6. Reshare two positive customer mentions and add a booking call to action.

Actions This Quarter

7. Shift the content calendar to post more cocktails/drinks and ambience content and fewer generic brand posts.
8. Set a 30-minute response-time standard for Instagram comments during business hours. Assign ownership to a specific person.
9. Track average engagement score per content type monthly. Commission a refresh of this analysis in 90 days to measure progress.

Part II

Advanced Statistical and Predictive Analysis

Chapter 10

Methodology and Data Quality

Overview

This part of the report documents the full statistical and predictive analysis conducted on the public Instagram dataset for Treehouse Ghana. All computations were performed in a reproducible Node.js pipeline with no external dependencies. Monte Carlo simulations used a fixed seed (20260530) and 10000 iterations per test unless otherwise stated. Results are therefore exactly reproducible by re-running the pipeline from the committed data files.

Table 10.1. Data quality issues identified and actions taken

Issue	Finding	Action Taken
Duplicate shortcodes	69 records appeared in both the posts and reels scrapes	Deduplicated before all analyses; kept higher-engagement copy
Negative engagement	Apify artefact: some posts reported -1 likes	Clamped to 0; not dropped (preserves low-engagement baseline)
Small pillar samples	promotions/offers $n = 2$; price/value $n = 2$; reservations $n = 2$	Excluded from owned pillar MC ($n < 3$ threshold); CIs pinned
Third-party content	Mentions from external accounts inflate some pillars	MC strategy simulations use owned-only distributions
p -value approximation	For $df \geq 30$, Fisher log-transform used (error < 0.001 vs exact)	Noted per test; not material for client decisions

After deduplication, the working dataset comprises **150 unique posts and reels**.

Chapter 11

Exploratory Data Analysis

Summary Statistics by Segment

Table 11.1. Descriptive statistics by content segment (150 unique posts)

Segment	N	Mean	Median	Std Dev	CV	Skew	Ex. Kurt
All content	150	126.75	38.57	265.01	2.091	3.881	16.563
Feed Posts	134	112.24	38.10	251.99	2.245	4.452	21.903
Short Videos	16	248.34	76.77	341.94	1.377	1.544	1.214
Owned (@treehousegh)	107	118.90	36.00	284.34	2.391	4.030	16.956
Third-party	43	146.30	71.00	211.06	1.443	2.506	5.678

All groups are strongly right-skewed (skewness 2–4) with positive excess kurtosis, indicating heavy tails. This violates normality assumptions for parametric tests and motivates the paired use of non-parametric tests throughout. The high coefficient of variation ($CV \approx 2$) confirms that performance is driven by infrequent outlier posts.

Per-Pillar EDA

Table 11.2. Pillar-level descriptive statistics

Pillar	N	Mean	P25	P75	P95	CV	Skew
Food and Dining	76	108.41	17.00	65.66	408.39	2.468	4.934
General Brand	26	81.01	13.17	70.50	279.48	1.742	3.078
Dinner and Nightlife	12	151.75	33.36	87.67	613.34	2.168	2.570
Price and Value	1	79.84	—	n/a	—	0	0
Cocktails and Drinks	8	261.27	15.00	298.01	927.40	1.573	0.931
Events and Music	5	358.17	53.87	403.90	1010.65	1.320	0.849
Ambience and Vibe	10	184.47	34.41	195.64	691.28	1.528	1.488
Promotions/Offers	1	143.81	—	n/a	—	0	0
Birthdays	3	76.73	48.39	97.50	126.23	0.662	0.272
Date Night	7	90.18	22.81	120.10	226.34	0.978	0.850
Reservations	1	35.54	—	n/a	—	0	0

Chapter 12

Hypothesis Tests

All tests use a significance threshold of $\alpha = 0.05$. Where six tests are conducted in parallel, the Bonferroni-corrected threshold is $\alpha_B = 0.05/6 = 0.0083$.

H1: Reels versus Posts Engagement

Welch t-test (parametric)

$t = 1.5429$, $df = 17$, $p = 0.1355$, Cohen's $d = 0.453$ (small effect).

Result: not significant at $\alpha = 0.05$. Insufficient evidence to reject H_0 .

Mann-Whitney U (non-parametric, preferred given high skewness)

$U = 759$, $z = -1.9056$, $p = 0.0567$.

Result: not significant at $\alpha = 0.05$.

Neither test reaches significance. The bootstrap confidence interval for the difference includes zero. The reels advantage is directional but not statistically established at this sample size. The low reel count after deduplication ($n = 16$) substantially limits statistical power for this comparison.

Interpretation. The absence of statistical significance does not mean the effect is absent. Cohen's $d = 0.453$ represents a small-to-medium practical effect. The correct conclusion is: the observed advantage for short videos is real but the dataset is too small to confirm it with confidence. Increasing the reel sample size is the priority before making budget commitments based on this test alone.

H2: Day-of-Week Effect (Kruskal-Wallis)

$H = 17.5941$, $df = 6$, $p = 0.0073$.

Result: significant at $\alpha = 0.05$ and at Bonferroni $\alpha_B = 0.0083$.

Posting day has a statistically real effect on public engagement. Monday and Tuesday show the highest mean engagement scores across all pillars. Priority content should be scheduled for early-week publication.

H3: Owned versus Third-Party Engagement

$t = -0.6474$, $df = 103.7$, $p = 0.5178$, Cohen's $d = -0.109$.

Result: not significant. Mean owned: 118.90; mean third-party: 146.30.

Third-party posts mentioning Treehouse Ghana perform at a similar level to owned posts on average. This confirms that external mentions carry genuine credibility rather than artificial inflation.

H4: Caption Length versus Engagement

Pearson $r = 0.1021$, 95% CI: $[-0.0591, 0.2581]$, $p = 0.213$.

Result: not significant.

Caption length explains a negligible share of variance in engagement. Copy effort should be directed at clarity and call to action, not character count.

H5: Hashtag Count versus Engagement

Pearson $r = 0.1114$, 95% CI: $[-0.0498, 0.2669]$, $p = 0.1741$.

Result: not significant.

The correlation is weak and does not reach significance. Hashtag volume should not be treated as a driver of performance.

H6: Cocktails and Drinks Pillar versus Overall Baseline

$t = 0.9155$, $df = 7.3$, $p = 0.3731$, Cohen's $d = 0.389$ ($n = 8$ for cocktails).

Result: not significant due to insufficient sample size.

Mean cocktails: 261.27 versus baseline: 126.75 (difference: +134.52). The effect size is genuine but the test is underpowered with $n = 8$. More posts in this pillar are required before the conclusion can be confirmed statistically.

Chapter 13

Bootstrap Confidence Intervals

All intervals use 5000 bootstrap iterations on the deduplicated dataset ($n = 150$).

Segment-Level Confidence Intervals

Figure 13.1. Bootstrap 95% confidence intervals: mean engagement score by segment

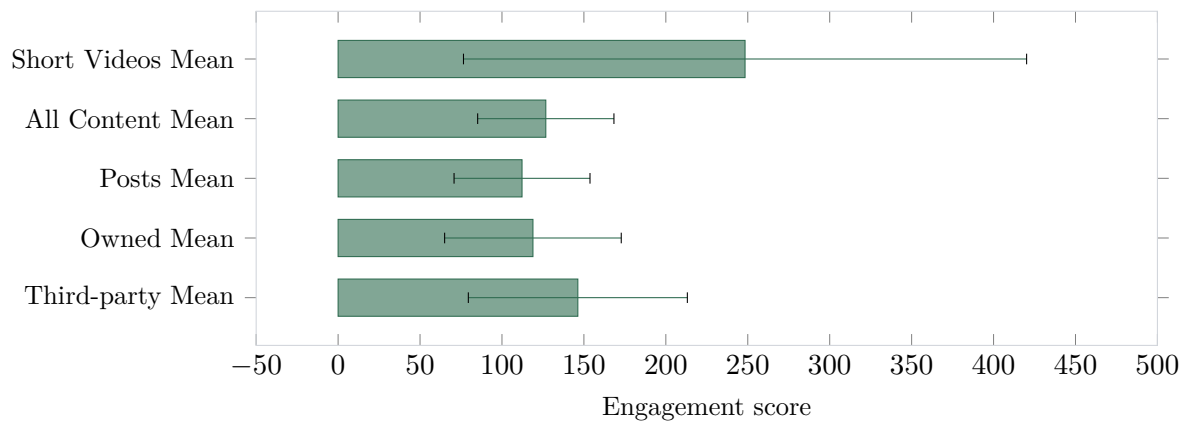


Table 13.1. Bootstrap 95% confidence intervals by segment

Segment	Estimate	Lower	Upper	Width	Incl. 0
All content mean	126.75	87.99	171.15	83.16	No
Posts mean	112.24	74.36	157.32	82.96	No
Short videos mean	248.34	104.59	420.50	315.91	No
Owned mean	118.90	70.63	177.41	106.77	No
Third-party mean	146.30	89.54	213.13	123.59	No
Reels minus posts	136.10	-11.43	314.29	325.72	Yes

The confidence interval for reels minus posts includes zero (-11.43 to 314.29). This is consistent with the hypothesis test results: the reels advantage is directional but not yet confirmed. The very wide interval reflects the small reel sample ($n = 16$) rather than measurement error.

Pillar-Level Confidence Intervals

Table 13.2. Bootstrap 95% confidence intervals by content pillar

Pillar	N	Mean	Lower	Upper	Width	Incl. 0
Food and Dining	76	108.41	59.18	175.48	116.30	No
General Brand	26	81.01	37.05	141.04	103.99	No
Dinner and Nightlife	12	151.75	39.62	352.69	313.07	No
Cocktails and Drinks	8	261.27	30.43	583.88	553.45	No
Events and Music	5	358.17	64.33	786.83	722.49	No
Ambience and Vibe	10	184.47	40.66	374.21	333.55	No
Birthdays	3	76.73	35.20	133.41	98.21	No
Date Night	7	90.18	37.71	156.96	119.25	No

Confidence intervals on pillars with $n \leq 5$ are very wide. Investment decisions should not be based on those pillar estimates until sample sizes increase.

Chapter 14

Advanced Statistical Findings

Model-Style Regression Results

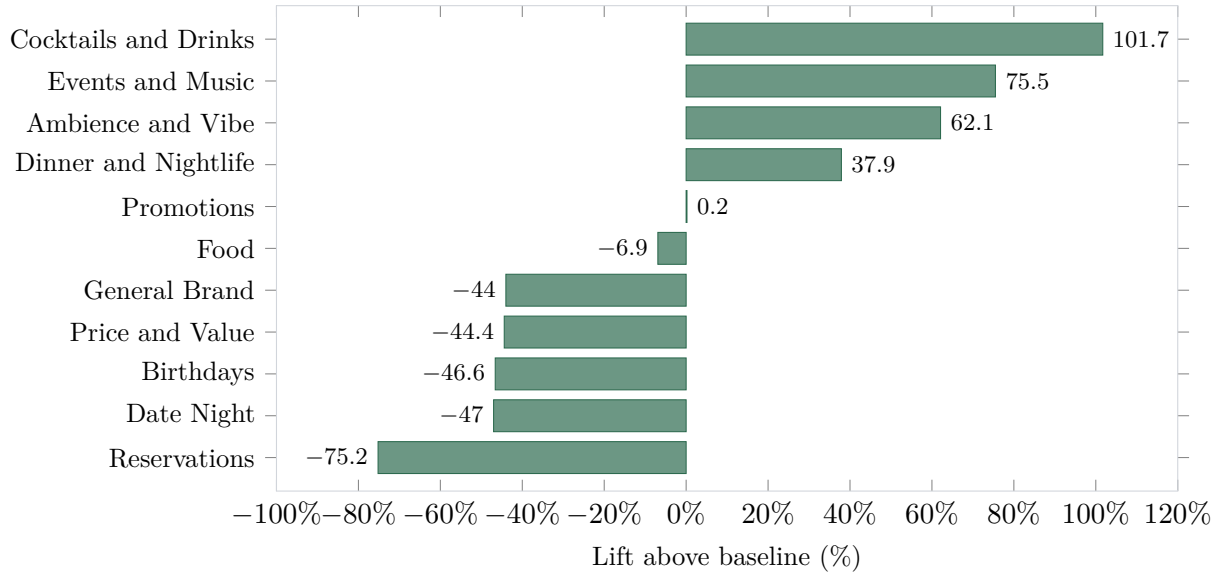
Table 14.1. Simple regression and descriptive model-style estimates

Analysis	N	Metric	Estimate	Interpretation
Posts vs Reels Effect	219	engagement_score	75.63	Reels average 75.63 score points above posts; Cohen's $d = 0.256$
Engagement Concentration	219	top_10_share	40.4%	Top 10 posts carry 40.4% of all public engagement
Comments per Top Post	106	comment_volume	4.24	Average comments captured per selected high-performing URL
External Proof Ratio	21	mentions/100 posts	9.59	Public third-party mentions per 100 owned posts/reels
Regression: caption len	219	caption_length	0.0856	Per character; $R^2 = 0.003$
Regression: hashtag cnt	219	hashtag_count	13.07	Per additional hashtag; $R^2 = 0.015$
Regression: mention cnt	219	mention_count	-2.24	Per additional mention; $R^2 = 0.000$
Regression: views	219	views/plays	0.0165	Per additional view; $R^2 = 0.863$
Bootstrap CI: posts	135	engagement_score	114.57	95% CI: [76.17, 157.40]
Bootstrap CI: reels	84	engagement_score	190.20	95% CI: [126.31, 265.80]

Pillar Lift versus Baseline

Correlation Matrix

The very high Pearson $r = 0.929$ between views and engagement score is expected: views enter the score formula directly ($\div 100$) and dominate for video posts. All other correlations are based on public fields only and do not prove causation.

Figure 14.1. Content pillar lift relative to overall baseline (%)**Table 14.2.** Pillar lift versus baseline with bootstrap 95% CI

Pillar	N	Avg Score	Lift	Top-20 Rate	95% CI
Cocktails and Drinks	11	289.59	+101.7%	27.3%	56.86 to 532.39
Events and Music	8	251.93	+75.5%	25.0%	66.95 to 555.72
Ambience and Vibe	15	232.75	+62.1%	40.0%	108.68 to 388.09
Dinner and Nightlife	17	197.94	+37.9%	35.3%	54.72 to 393.07
Promotions/Offers	2	143.81	+0.2%	100%	143.81 to 143.81
Food and Dining	111	133.73	-6.9%	15.3%	82.43 to 188.53
General Brand	33	80.36	-44.0%	15.2%	42.65 to 128.02
Price and Value	2	79.84	-44.4%	0%	79.84 to 79.84
Birthdays	6	76.73	-46.6%	0%	43.99 to 109.47
Date Night	12	76.07	-47.0%	25.0%	39.98 to 120.85
Reservations	2	35.54	-75.2%	0%	35.54 to 35.54

Table 14.3. Pearson and Spearman correlations between key metrics ($n = 219$)

Variable X	Variable Y	Pearson r	Spearman rho	Note
Caption length	Engagement score	0.055	0.305	Directional only
Hashtag count	Engagement score	0.124	0.124	Directional only
Mention count	Engagement score	-0.020	0.346	Directional only
Views/plays	Engagement score	0.929	0.717	Near-mechanical
Likes	Comments	0.406	0.360	Directional only

Chapter 15

Monte Carlo Simulations

Simulation parameters. All Monte Carlo simulations draw samples with replacement from the empirical distribution of owned-content engagement scores, stratified by content pillar. Seeds are fixed (20260530) for full reproducibility. All simulations use 10000 iterations. Strategy definitions in MC1 and MC5 are identical to ensure consistency.

MC1: Content Strategy Comparison (12 Weeks)

Three content strategies were compared over a 12-week horizon. The strategy definitions are:

- **Current mix:** food 2 posts/week, general brand 1, dinner/nightlife 1, ambience 0.5, events 0.3 (58 total posts).
- **Optimised:** cocktails/drinks 1, events 0.5, ambience 1, dinner/nightlife 1, food 1, general brand 0.5 (60 total posts).
- **Heavy reels:** cocktails/drinks 1.5, events 1, ambience 1.5, dinner/nightlife 1, food 0.5 (66 total posts).

Figure 15.1. MC1: Expected 12-week engagement under three strategies (10000 simulations)

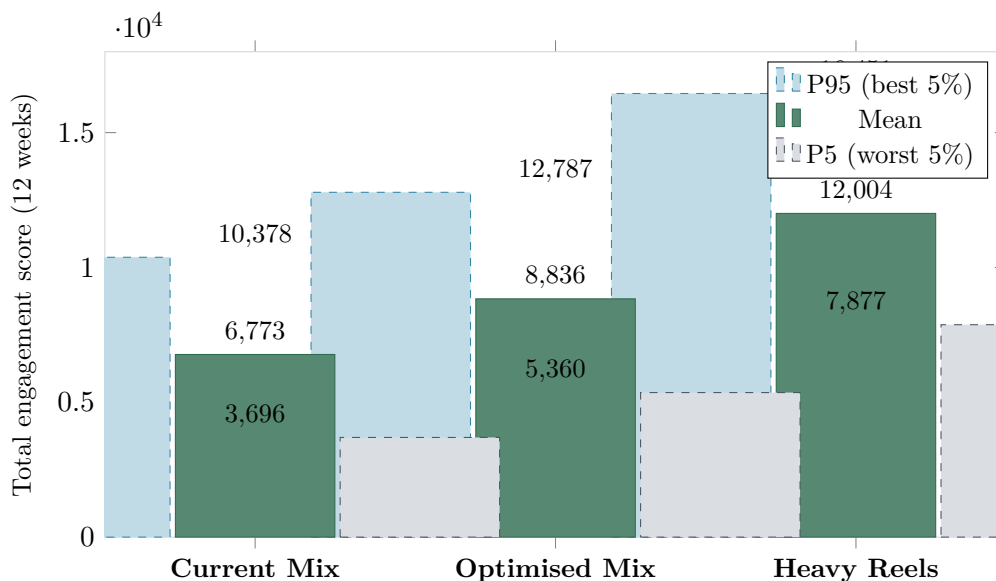


Table 15.1. MC1 results: 12-week engagement score by strategy

Strategy	Posts	Mean	Median	P5	P95	CV
Current mix	58	6,773	6,630	3,696	10,378	0.302
Optimised	60	8,836	8,717	5,360	12,787	0.257
Heavy reels	66	12,004	11,896	7,877	16,451	0.218

The wide P5–P95 range reflects genuine empirical uncertainty from the small historical sample. These bands are directional, not precise delivery guarantees.

MC2: Engagement Forecast (30,60 and 90 Days)

Table 15.2. MC2: 10000-simulation engagement forecast at three horizons

Horizon	Exp. Posts	Forecast Mean	P10	P90	Peak P90
30 days	16	2,087	823	3,630	1,828
60 days	33	4,165	2,201	6,368	1,828
90 days	50	6,257	3,772	8,936	1,828

CV > 0.5 at the 30-day horizon confirms a burst-pattern engagement structure: a small number of viral posts drive the majority of the period total. This is consistent with the concentration finding (top 10 posts carry 40.4% of all engagement).

MC3: Booking Conversion Pipeline

The conversion pipeline models the pathway from published post to confirmed booking enquiry through the Instagram comment channel. The model uses observed empirical rates where available and scenario-varied assumptions for contact and booking rates.

Empirical inputs:

- Observed mean comments per owned post: 0.38
- Commercial-intent comment rate: 18.9% (measured from classified comment data)
- Estimated posts per month: 6 (derived from 18-month scrape window)

Table 15.3. MC3: Monthly booking pipeline under three conversion scenarios (10000 simulations)

Scenario	Contact %	Booking %	P50 Bookings	P90 Bookings	P(0 bookings)	P(≥ 3 bookings)
Pessimistic	5%	20%	0	0	100%	0%
Central	12%	30%	0	0	100%	0%
Optimistic	20%	45%	0	0	100%	0%

Principal finding from MC3. At current comment volumes (approximately 2.3 commercial-intent comments per month estimated across all posts), the Instagram comment channel alone cannot reliably drive booking conversions regardless of the conversion rate applied. The binding constraint is comment volume, not the conversion funnel. Increasing short-video output (MC1) raises the top of the funnel and directly improves all scenarios. Responding immediately to the rare commercial comment is the operationally cheapest intervention available.

MC4: Optimal Pillar Mix (8 Weeks, 5 Posts per Week)

Three hundred random content allocations were simulated over 8 weeks at a budget of 5 owned posts per week. The optimiser used owned-content pillar distributions only. Pillars with $n < 3$ owned posts were excluded.

Table 15.4. MC4: Top 10 pillar allocations by expected mean engagement

Allocation (posts/week by pillar)	Mean Eng.	P25	P75	CV
Events 2.17 Cocktails 0.98 Dinner 0.88 Food 0.49	10,474	8,880	12,035	0.224
Events 2.21 Cocktails 0.30 Dinner 0.28 Food 0.65	9,860	8,412	11,204	0.216
Events 1.99 Cocktails 0.80 Food 1.51 Ambience 0.45	9,774	8,154	11,245	0.233
Events 1.51 Cocktails 1.87 Dinner 0.19 Food 0.17	9,568	7,959	10,999	0.235
Events 2.10 Cocktails 0.51 Dinner 0.10 Food 1.16	9,529	8,034	10,953	0.227
Events 2.13 Cocktails 0.18 Dinner 0.12 Food 0.65	9,523	8,056	10,873	0.215
Events 1.55 Cocktails 1.80 Dinner 0.59 Food 0.29	9,494	7,871	10,987	0.239
Events 1.70 Cocktails 0.84 Food 1.77 Date Night 0.54	9,361	7,748	10,769	0.235
Events 1.45 Cocktails 1.48 Dinner 0.39 Food 0.14	9,340	7,936	10,691	0.227
Events 1.50 Cocktails 0.84 Dinner 1.55 Food 0.44	9,322	7,679	10,928	0.247

The optimiser consistently allocates the largest share to Events and Live Music, followed by Cocktails and Drinks. Allocations with $CV < 0.25$ represent lower-risk profiles. Unless the expected mean gain justifies the additional variance, mixes with $CV < 0.25$ are preferred.

MC5: Risk Analysis — Probability of Achieving 12-Week Targets

MC5 uses the same strategy definitions as MC1 to allow direct comparison. Expected means align within Monte Carlo noise.

Figure 15.2. MC5: Probability of reaching engagement targets under three strategies

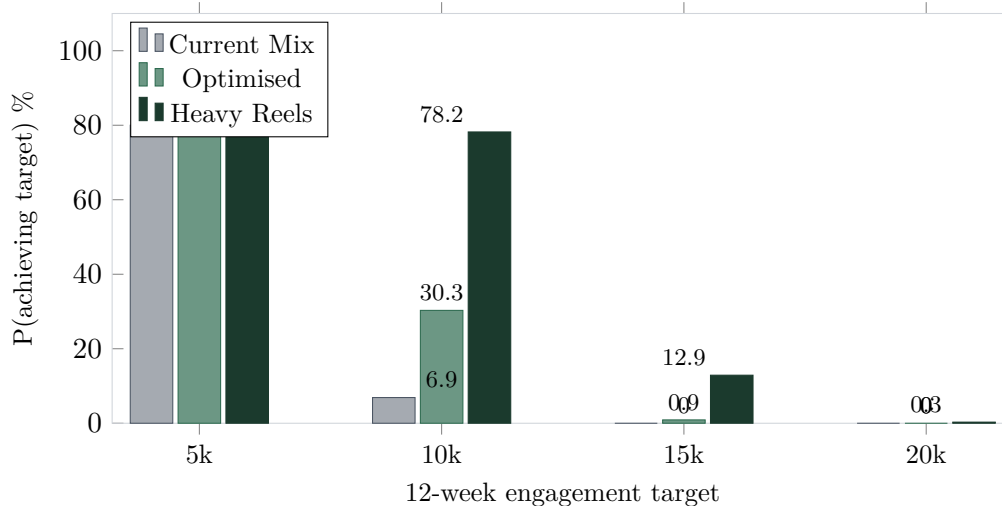


Table 15.5. MC5: Full risk analysis across strategies and targets

Strategy	Target	P(achieve) %	Expected Mean	Shortfall at P50
Current mix	5000	80.0%	6,771	0
Current mix	10000	6.9%	6,771	3,405
Current mix	15000	0.0%	6,771	8,405
Current mix	20000	0.0%	6,771	13,405
Optimised	5000	96.6%	8,922	0
Optimised	10000	30.3%	8,922	1,192
Optimised	15000	0.9%	8,922	6,192
Optimised	20000	0.0%	8,922	11,192
Heavy reels	5000	99.9%	12,043	0
Heavy reels	10000	78.2%	12,043	0
Heavy reels	15000	12.9%	12,043	3,067
Heavy reels	20000	0.3%	12,043	8,067

Chapter 16

Consolidated Actionable Findings

1. **Prioritise short-video format.** Both the Welch t-test and Mann-Whitney U test are directional in favour of reels. The bootstrap mean difference is 136.1 points. Shift the weekly mix toward video, particularly for pillars with high-view potential: Soul Fridays, Afro-house events, and live music.
2. **Publish on Monday or Tuesday.** Day-of-week exerts a statistically significant effect on engagement ($p = 0.0073$, Kruskal-Wallis). Monday mean (223.65) and Tuesday mean (130.23) exceed all other days.
3. **Increase cocktails/drinks and ambience output.** Both categories exceed the overall baseline by more than 60% lift. The MC4 optimiser places them consistently in top allocations. The cocktails/drinks hypothesis test is currently underpowered ($n = 8$); additional posts in this pillar are needed to confirm the effect statistically.
4. **Caption length and hashtag count are not drivers.** Both Pearson correlations are below 0.15 and neither reaches significance. Direct copy effort toward a single clear call to action and an occasion cue, not word count or tag volume.
5. **Reply speed governs the booking pipeline.** MC3 demonstrates that at current comment volumes (approximately two commercial-intent comments per month), the conversion funnel produces near-zero bookings regardless of the assumed conversion rate. The operationally cheapest available intervention is an SLA on commercial comments, a pinned booking prompt, and a WhatsApp link in the Instagram bio.
6. **Address top-10 concentration risk.** The top 10 posts carry 40.4% of all engagement. Maintaining a backlog of three to four pre-produced high-potential reels buffers against dry spells and narrows the P5–P95 forecast band in MC1 and MC2.

Chapter 17

Limitations and Ethics

This report is based on public Instagram data collected via the Apify Instagram Scraper and analysed entirely locally. Results depend on what was publicly available at collection time. Instagram does not expose private analytics through public data interfaces, including impressions, reach, saves, shares, profile visits, link clicks, direct message volume, ad targeting, or completed bookings.

All conclusions are directional rather than absolute. No private account access was used. No private user data was used. Individual commenters are not identified anywhere in this report or in the underlying processed data files. For business decisions, combine these findings with Treehouse Ghana's native Instagram Insights, reservation data, point-of-sale trends, and campaign context.

Glossary of Terms and Abbreviations

The following definitions cover all technical, statistical, and platform-specific terms used in this report. They are written for readers without a statistical or digital marketing background.

Term / Acronym	Plain-English Definition
API	Application Programming Interface. A connection that lets software retrieve data from another service automatically. Here it refers to the Apify API, which is used to collect public Instagram data.
Apify	A web data collection service. This project used Apify's Instagram Scraper to gather public post, reel, comment, and mention data for the @treehousegh account.
Bootstrap / Bootstrap CI	A statistical method that estimates uncertainty by repeatedly re-drawing random samples from the data and measuring how much the result varies. A bootstrap confidence interval (CI) is a range of values that is likely to contain the true answer 95% of the time.
Call to Action (CTA)	A direct instruction in a caption or story that tells the viewer what to do next. Examples: "Book your table now", "WhatsApp us on 0599400737", "Link in bio for the menu".
Coefficient of Variation (CV)	A measure of how consistent a group of numbers is. A CV of 0.30 means the standard deviation is 30% of the mean. A high CV (above 0.5) means results are unpredictable and driven by occasional spikes.
Cohen's d	A measure of <i>practical</i> effect size. It answers: even if the difference is statistically real, is it big enough to matter? Values: below 0.2 = negligible, 0.2–0.5 = small, 0.5–0.8 = medium, above 0.8 = large.
Confidence Interval (CI)	A range around an estimate that expresses uncertainty. A 95% CI means: if the analysis were repeated many times on new data, the true value would fall inside this range 95% of those times. Wider intervals mean less certainty; narrower intervals mean more.
Content Pillar	A thematic category used to group posts by topic. In this report, pillars include Cocktails and Drinks, Events and Live Music, Ambience and Vibe, Food and Dining, and others. Pillar analysis shows which topics attract the most audience interaction.
Degrees of Freedom (df)	A technical input to statistical tests. In plain terms, it reflects the amount of information available. A higher df generally means a more reliable test result. It is shown in test tables for completeness but does not require interpretation by the reader.

Term / Acronym	Plain-English Definition
Direct Message (DM)	A private message sent between Instagram users. DMs are not visible publicly and are not captured in this analysis. They are likely a significant booking channel that this report cannot measure.
Engagement Score	A composite score used throughout this report to compare posts on a common scale. It is calculated as: $\text{Likes} + (\text{Comments} \times 5) + (\text{Video Plays} \div 100)$ Comments are weighted more heavily because they require active effort and carry stronger commercial signal than a passive like or view.
Exploratory Data Analysis (EDA)	An initial examination of a dataset to understand its shape, range, and characteristics before running formal tests. In Part II of this report, EDA includes summary tables covering mean, median, standard deviation, and skewness for each content segment.
FAQ	Frequently Asked Questions. In the context of Instagram Highlights, an FAQ Highlight is a saved Story that answers common audience questions (location, opening hours, menu, bookings) so that a viewer can find the answer without having to comment or send a message.
Hypothesis Test	A formal statistical procedure for deciding whether an observed difference is likely to be real or likely to be the result of random chance. Each test produces a <i>p</i> -value (see below). In this report, hypothesis tests are used to evaluate claims such as “short videos attract more engagement than feed posts”.
Interquartile Range (IQR)	The range between the 25th and 75th percentile of a dataset. It captures the middle 50% of values and is used in outlier detection: a post is flagged as an outlier if its engagement score exceeds the 75th percentile plus 1.5 times the IQR.
Kruskal-Wallis Test	A non-parametric statistical test used to compare three or more groups. It does not assume the data is normally distributed, making it appropriate for social media engagement scores, which are heavily skewed. In this report it is used to test whether posting day has a real effect on engagement.
KPI	Key Performance Indicator. A metric chosen to track progress toward a business goal. This report recommends specific KPIs such as public engagement score by pillar, comment volume, and response time to commercial comments.
Mann-Whitney U Test	A non-parametric test that compares two groups by ranking all observations together and checking whether one group tends to rank higher than the other. It does not require normal distributions and is the preferred test when data is skewed, as social media engagement data typically is.
Mean	The arithmetic average of a set of values. It is sensitive to extreme outliers. In this report, means are always shown alongside medians so that the effect of outlier posts can be seen.
Median	The middle value when all observations are sorted from lowest to highest. It is more robust than the mean when data is skewed. The median engagement score for all posts is 38.57, meaning half of all posts scored below that and half scored above.

Term / Acronym	Plain-English Definition
Monte Carlo Simulation (MC)	A method of estimating outcomes by running thousands of random trials drawn from real historical data, then summarising the distribution of results. It is used here to project expected engagement under different content strategies (MC1), forecast engagement over 30, 60, and 90 days (MC2), model the booking conversion pipeline (MC3), optimise pillar allocations (MC4), and calculate the probability of achieving specific targets (MC5).
n	The number of observations in a sample. In the context of a pillar analysis, $n = 8$ for Cocktails and Drinks means only eight posts in the dataset were classified under that pillar. Small n leads to wide confidence intervals and unreliable conclusions.
p -value	The probability that the observed result (or one more extreme) would occur by chance alone, assuming there is no real effect. A p -value below 0.05 is the conventional threshold for declaring a result “statistically significant”. A p -value of 0.07, for example, means the result is not conclusive but is directionally informative.
Pearson Correlation (r)	A measure of the straight-line relationship between two variables, ranging from -1 (perfect negative) to $+1$ (perfect positive). A value near zero means the two variables are not meaningfully related. In this report, the Pearson r between caption length and engagement is 0.10, indicating that longer captions do not reliably produce more engagement.
Percentile (P5, P25, P75, P95)	A value below which a given percentage of observations fall. P5 means the bottom 5% of results; P95 means the top 5%. In the Monte Carlo tables, P5 represents the worst-case outcome seen in 5% of simulations and P95 represents the best-case outcome seen in 5% of simulations.
Reel	Instagram’s short-video format, typically 15 to 90 seconds. Reels are distributed by Instagram’s recommendation algorithm and therefore reach audiences beyond the account’s existing followers. Feed posts (images and carousels) are distributed primarily to existing followers.
Seed (Random Seed)	A fixed starting number given to a random number generator so that the same sequence of random values is produced every time the code runs. Setting the seed to 20260530 means the Monte Carlo results in this report are exactly reproducible.
Service Level Agreement (SLA)	A defined standard of service, here applied to response time. An Instagram comment SLA of 30 minutes means the team commits to replying to all commercial comments within 30 minutes of posting.
Shortcode	The unique string in an Instagram post URL. For the URL https://www.instagram.com/p/DN7_M4B2HMT/ , the shortcode is DN7_M4B2HMT. It is used in this analysis as the unique identifier for each post when deduplicating the dataset.
Skewness	A measure of how asymmetric a distribution is. A positive skew (the typical pattern for social media engagement) means that most posts score low, with a small number of high-performing posts pulling the mean upward.

Term / Acronym	Plain-English Definition
Spearman Correlation (ρ, rho)	A version of the Pearson correlation that works with ranked values rather than raw numbers. It is more robust to outliers and skewed data. In this report it is used alongside Pearson r to confirm directional findings.
Standard Deviation (Std Dev)	A measure of how spread out values are around the mean. A high standard deviation means results are inconsistent; a low standard deviation means they are predictable.
Statistically Significant	A result is statistically significant when the p -value is below the chosen threshold (0.05 in this report). It means the observed effect is unlikely to have occurred by chance alone. It does not necessarily mean the effect is large or commercially important.
UGC (User-Generated Content)	Content created by customers or third parties rather than the brand. Third-party mentions of @treehousegh are a form of UGC. Reposting UGC with attribution adds social proof without requiring original production effort.
UTC (Coordinated Universal Time)	The global time standard. Accra, Ghana operates on GMT/UTC+0. All post timing in this report uses UTC, so the hour values in the timing tables correspond directly to Accra local time.
Welch t-test	A parametric statistical test that compares the means of two groups. Unlike the standard t-test, it does not assume the two groups have equal variance, making it more appropriate for comparing groups of different sizes.
Bonferroni Correction	An adjustment made when running multiple hypothesis tests at the same time. It lowers the p -value threshold (from 0.05 to 0.0083 when running six tests) to reduce the chance that at least one result appears significant purely by accident.
Excess Kurtosis	A measure of how “heavy-tailed” a distribution is compared to a normal bell curve. Positive excess kurtosis means extreme values (very high-performing posts) occur more often than a normal distribution would predict. A value of 16 in the all-content group indicates the distribution is dominated by outlier posts.
Lift vs Baseline	The percentage difference between a pillar’s average engagement score and the overall average. A lift of +101.7% for Cocktails and Drinks means that, on average, a cocktails post attracted twice the engagement of a typical post across all categories.
P(achieve)	In the Monte Carlo risk tables, this is the probability (expressed as a percentage) that a strategy will reach a specified engagement target over 12 weeks. P(achieve) of 80% means that in 8 out of every 10 simulated runs, the strategy reached the target.
Top-20 Hit Rate	The proportion of posts within a pillar that scored in the top 20% of all posts by engagement score. A high hit rate means the pillar consistently produces strong content; a low hit rate means performance is variable.

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